

PROGRAMMING FUNDAMENTALS

Spring 2024

6th March

Increment & Decrement Operators

- num++
- num--
- ++num
- --num

```
num = 4;  
cout << num++;
```

```
num = 4;  
cout << ++num;
```

Increment & Decrement Operators

- `x = 10;`
 - `if (x++ > 10)`
 - `cout << "x is greater than 10 \n";`
-
- `x = 10;`
 - `if (++x > 10)`
 - `cout << "x is greater than 10 \n";`
-
- page 801 (tony Gaddis) → try these

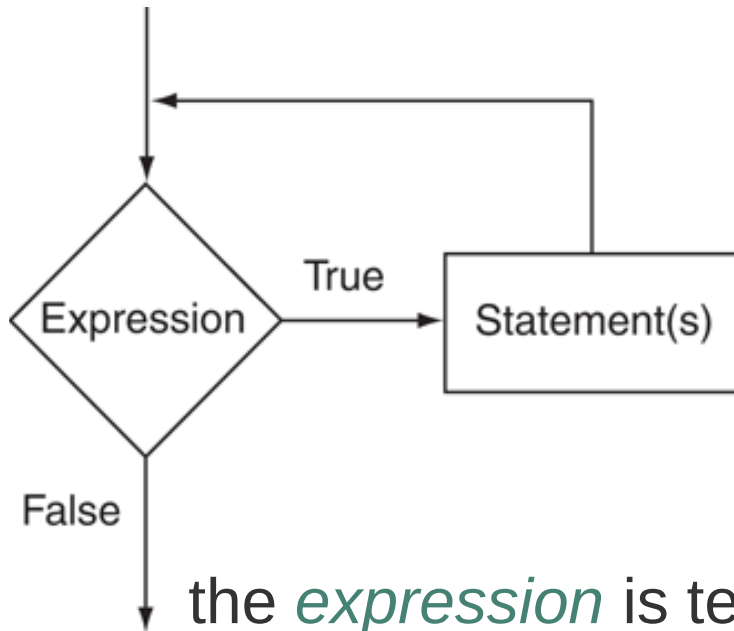
Loop

Repeating a set of statements for a set number of times. Three types in C++

1. WHILE Loop
2. Do..While Loops
3. FOR Loop

WHILE Loop - Format

1. Loop Variable
2. Condition in WHILE statement
3. Loop Body to repeat
4. Increment/decrement statement for loop variable



```
while (expression)  
statement;
```

the *expression* is tested, and if it is true,
the *statement* is executed

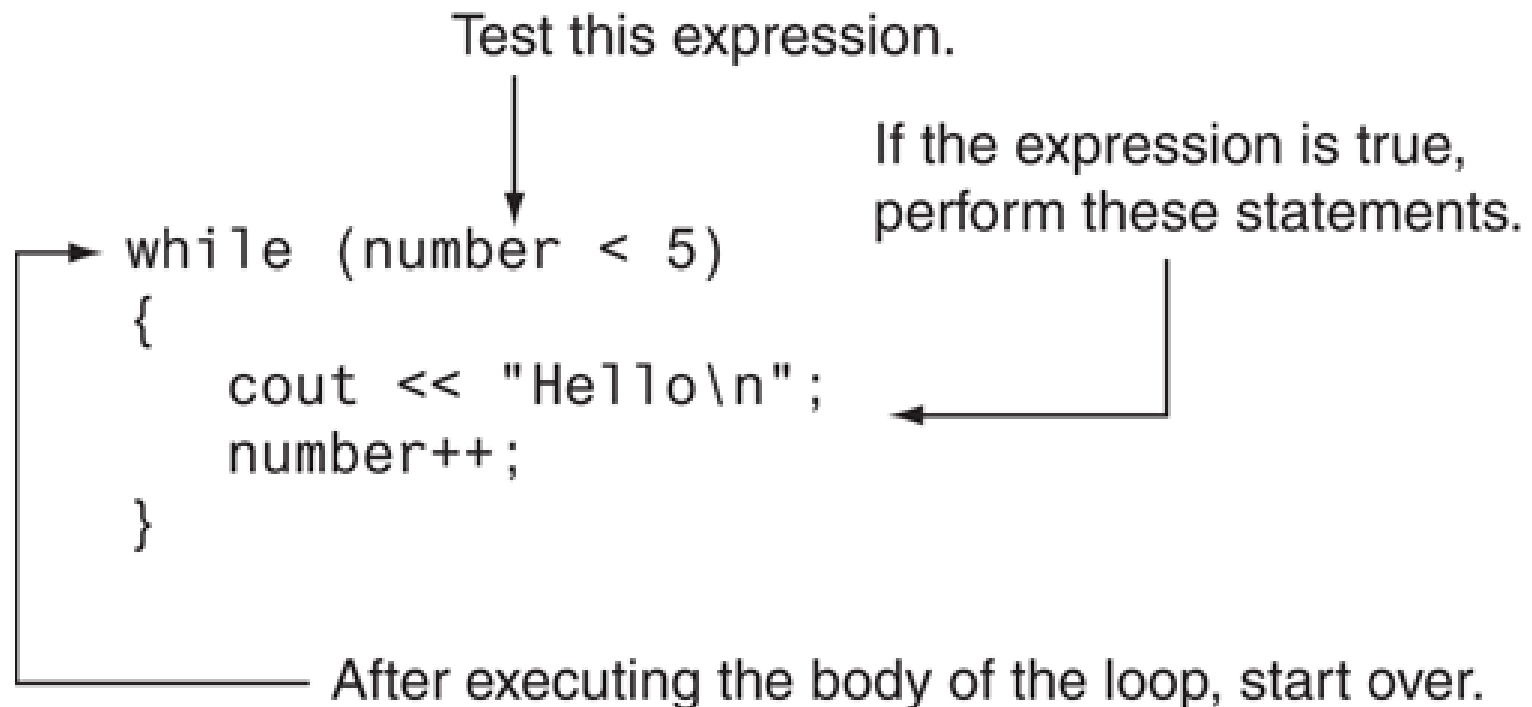
Then, the *expression* is tested again. If it
is true, the *statement* is executed.

This cycle repeats until the *expression* is false

Running while Loop

- Like the if statement, the while loop is not complete without the statement that follows it
- Known as a *pretest* loop, which means it tests the expression before each iteration
- Loop will never iterate if the test expression is false to start with
- If you want to be sure that a while loop executes the first time, you must initialize the relevant data in such a way that the test expression starts out as *true*
- If a loop does not have a way of stopping, it is called an infinite loop

Running the while Loop



Ex-1

Write a code to print first 7 integers (1..7) using WHILE loop

Ex-2

Write a code to print the sum of first 7 integers (1..7) using WHILE loop

Ex-3

- Ask the user to enter an integer value, x
- Your loop should run x number of times
- Accept x inputs
- Add all entered numbers
- Display the final sum

Ex-4

Write code to display the square of first five integers using WHILE loop structure:

Square = 1

Square = 4

Square = 9

Square = 16

Square = 25

Ex-5

Modify your code to display first five integers along with their squares using WHILE loop structure

n = 1	Square = 1
n = 2	Square = 4
n = 3	Square = 9
n = 4	Square = 16
n = 5	Square = 25

Ex - 6

Write a program to print counting from 10 to 1 (in reverse order) using WHILE loop

Ex-7

Ask the user to enter a number. Write code to display the table of the given number in the following format using WHILE loop:

$$2 * 1 = 2$$

$$2 * 2 = 4$$

$$2 * 3 = 6$$

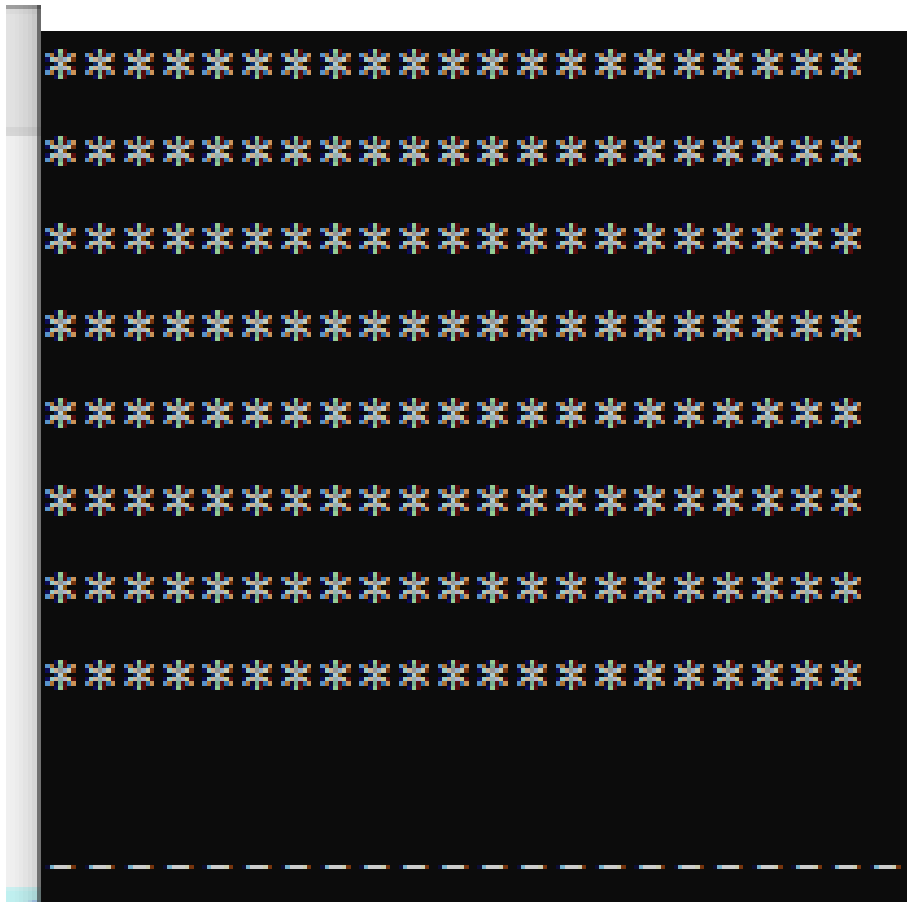
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Patterns while Loops



Printing 20 *s
with loop on 8
rows

Ex-8

- Modify example-3 to accept x number of inputs from the user but your code should only add the positive numbers entered. Any negative numbers will not be added to the sum.

Ex-9

- Convert the code from example-5 into an infinite loop

Ex-10

- Ask the user to enter an integer and a letter. Your code (using while loop should run only if a positive number is entered and character 'Y' is entered. Otherwise it should print "Maybe next time....."

Book Reference

- D.S. Malik (ch - 4)
- Tony Gaddis (ch - 4)